We observe that

a) The variance of $\widehat{\beta_2}$ is directly proportional to σ^2 and inversely proportional to $\sum x_i^2$. Thus, if the variation in *X* values increases, the value of $\sum x_i^2$ will increase. Therefore, the X value should be widely dispersed. In other words, β_2 can be measured with greater accuracy if there is considerable variation in *X* values.

b) As sample size *n* increases, the value of $\sum x_i^2$ will increase. Therefore, it is useful to have a large sample.

c) The variance of $\widehat{\beta_1}$ is directly proportional to σ^2 and $\sum X_i^2$. However, it is inversely proportional to $\sum x_i^2$ and the sample size *n*.

The estimators $\widehat{\beta_1}$ and $\widehat{\beta_2}$ will not just vary from one sample to another. The estimates $\widehat{\beta_1}$ and $\widehat{\beta_2}$ be dependent on each other in a sample. Such dependency is measured by the covariance between them.

4.3.2 Estimator of Error Variance

The variance of $\hat{\beta}_1$ and $\hat{\beta}_2$ depends on σ^2, he variance of u_i. The error variance (that is, the variance of u_i), however, is not known to us. The stimator of σ^2 can be obtained by OLS method and the aximum likelihood method. The OLS estimator $\widehat{\sigma}^2$ is given by

$$\widehat{\sigma}^2 = \frac{\sum \hat{u}_i^2}{n-2} = \frac{RSS}{n-2} \qquad \text{... (4.44)}$$

Note that $\widehat{\sigma}^2$ is the OLS estimator of the true but unknown σ^2 with $(n - 2)$ degrees of freedom. Once $\sum \hat{u}_i^2$ is estimated as the sum of squares of the residuals (RSS) given at equation (4.3), we can calculate $\widehat{\sigma}^2$ by using (4.44).

In the maximum likelihood (ML) method we obtained the estimator $\tilde{\sigma}^2 = \frac{RSS}{n}$. Thus, the ML estimator of σ^2 is biased. However, it is consistent, in the sense that as we increase the sample size infinitely, $\tilde{\sigma}^2$ approaches the true value of the parameter σ^2.

4.4 PROPERTIES OF THE OLS ESTIMATORS

The properties of OLS estimators can be classified as: (i) algebraic or numerical, and (ii) statistical. The statistical properities pertain to unbiasedness, consistency and efficiency of the OLS estimators. The algebraic properties of OLS estimators pertain to the relationship among the variables in the regression model.

4.4.1 Algebraic Properties

The algebraic properties of OLS estimators are as follows:

(i) The OLS estimators provide point estimates. Thus, *each estimator provides a single value of the population parameter*.

(ii) The sample regression line passes through the sample means of *Y* and *X*. The estimated regression line is given by $\hat{Y}_i = \widehat{\beta_1} + \widehat{\beta_2}X_i$. If we add up over the sample, and divide it by *n*, we obtain $\overline{Y} = \widehat{\beta_1} + \widehat{\beta_2}\bar{X}$. Thus, when X takes the value $\bar{X}$, the dependent variable Y takes the value $\overline{Y}$. The regression line passes through the point $(\bar{X}, \overline{Y})$

(iii) For the dependent variable, the mean of actual the actual value (Y) is equal to the mean of the estimated value $(\hat{Y})$. If we substitute the value of $\widehat{\beta_1}$ from equation (4.13) in (4.2), we obtain

$$\hat{Y}_i = \widehat{\beta_1} + \widehat{\beta_2}X_i$$
$$= (\overline{Y} - \widehat{\beta_2}\overline{X}) + \widehat{\beta_2}X_i = \overline{Y} - \widehat{\beta_2}(X_i - \overline{X})$$

Let us add up both the sides of equation (4.38), and divide by the sample size *n*. Further, let us apply the fact that $\sum(X_i - \overline{X}) = 0$. This gives us

$$\overline{\hat{Y}} = \overline{Y} \quad \text{... (4.45)}$$

(iii) The sample regression function can be expressed in deviation form as $\hat{y}_i = \widehat{\beta_2}x_i$.

We know that $\sum \hat{u}_i = 0$.

$$\hat{Y}_i = \widehat{\beta_1} + \widehat{\beta_2}X_i \quad \text{… (4.2)}$$

and

$$Y_i = \left(\hat{Y}_i + \hat{u}_i\right) = (\widehat{\beta_1} + \widehat{\beta_2}X_i) + \hat{u}_i$$

Or,

$$Y_i = \widehat{\beta_1} + \widehat{\beta_2}X_i + \hat{u}_i \quad \text{… (4.3)}$$

Summing up both the sides of equation (4.2) and dividing by the sample size *n*, we obtain

$$\overline{\hat{Y}} = \widehat{\beta_1} + \widehat{\beta_2}\overline{X}$$

Since $\overline{\hat{Y}} = \overline{Y}$, we have

$$\overline{Y} = \widehat{\beta_1} + \widehat{\beta_2}\overline{X} \quad \text{... (4.46)}$$

Subtracting (4.43) from (4.3) we get

$$(Y_i - \overline{Y}) = \widehat{\beta_2}\left(X_i - \overline{X}\right) + \hat{u}_i \quad \text{… (4.47)}$$

If we define y_i and x_i as deviations from their sample means, the above can be expressed as

$$y_i = \widehat{\beta_2}x_i + \hat{u}_i \quad \text{... (4.48)}$$

From (4.2) above, we have

$\overline{Y} = \widehat{\beta_1} + \widehat{\beta_2}\overline{X}$

Subtracting the above from (4.2), we obtain the regression line in deviation form as

$\widehat{y_i} = \widehat{\beta_2} x_i$ … (4.49)

(iv) The predicted value $\widehat{y_i}$ and the residual $\hat{u}_i$ are not correlated. It implies, we have to prove that $E(\hat{y}_i \hat{u}_i) = 0$, or $\frac{1}{n}\sum \widehat{y_i}\,\hat{u}_i = 0$.

From (4.44) $\widehat{y_i} = \widehat{\beta_2} x_i$. If we multiply both the sides by $\widehat{u_i}$ and take summation over all the observations, we get:

$$\sum \widehat{y_i}\,\hat{u}_i = \widehat{\beta_2} \sum x_i\,\hat{u}_i$$

$$= \widehat{\beta_2} \sum x_i\,(y_i - \widehat{\beta_2} x_i)$$

$$= \widehat{\beta_2} \sum x_i\, y_i - \widehat{\beta_2}^2 \sum x_i^2$$

$$= \widehat{\beta_2} \cdot \frac{\widehat{\beta_2}}{\widehat{\beta_2}} \sum x_i\, y_i - \widehat{\beta_2}^2 \sum x_i^2$$

$$= \widehat{\beta_2} \cdot \frac{\widehat{\beta_2}}{\widehat{\beta_2}} \sum x_i\, y_i - \widehat{\beta_2}^2 \sum x_i^2$$

$$= \widehat{\beta_2}^2 \cdot \frac{\sum x_i^2}{\sum x_i y_i} \cdot \sum x_i\, y_i - \widehat{\beta_2}^2 \sum x_i^2 = 0$$

$$= \widehat{\beta_2}^2 \sum x_i^2 - \widehat{\beta_2}^2 \sum x_i^2 = 0 \quad \text{... (4.50)}$$

Hence, the residuals $\widehat{u_i}$ and Y_i are not correlated.

(v) The residual $\widehat{u_i}$ and X_i are not ncorrelated. This implies, $E(X_i \widehat{u_i}) = 0$. In other words, $\frac{1}{n}\sum X_i\,\widehat{u_i} = 0$.

While taking the partial derivative of $\sum \widehat{u_i}^2$ with respect to $\widehat{\beta_2}$ (see equation (4.11)) we find that

$$\frac{\partial \sum \widehat{u_i}^2}{\partial \widehat{\beta_2}} = \frac{\partial \sum (Y_i - \widehat{\beta_1} - \widehat{\beta_2} X_i)^2}{\partial \widehat{\beta_2}} = 0$$

Or, $2 \cdot (-1) \cdot \sum (Y_i - \widehat{\beta_1} - \widehat{\beta_2} \mathrm{X_i}) \cdot X_i = 0$

Or, $\sum X_i (Y_i - \widehat{\beta_1} - \widehat{\beta_2} \mathrm{X_i}) = 0$

Or, $\sum X_i \widehat{u_i} = 0$ ... (4.51)

4.4.2 Statistical Properities

The OLS estimators fulfil certain statistical or theoretical properties. We specify them below.

(i) **Unbiasedness**

The OLS estimators are unbiased.

$$E(\widehat{\beta_2}) = E\left(\frac{\sum x_i y_i}{\sum x_i^2}\right) = E\left(\frac{\sum x_i(\beta x_i + u_i)}{\sum x_i^2}\right) = E\left(\frac{\sum \beta x_i^2 + \sum x_i u_i)}{\sum x_i^2}\right)$$

$= \beta + E\left(\frac{\sum x_i u_i}{\sum x_i^2}\right) = \beta + \left(\frac{\sum x_i E(u_i)}{\sum x_i^2}\right) = \beta + 0 = \beta$

Thus, $E(\widehat{\beta_2}) = \beta_2$ … (4.52)

(ii) **Consistency**

The OLS estimators are consistent. As we increase the sample size, $\widehat{\beta_2}$ converges to β_2.

$\widehat{\beta_2} = \frac{\sum x_i y_i}{\sum x_i^2} = \beta + \frac{\sum x_i u_i}{\sum x_i^2} = \beta + \frac{\sum x_i u_i / n}{\sum x_i^2 / n}$ … (4.53)

As $n \to \infty$, we observe that the covariance between X and u is zero, that is, $\sum x_i u_i / n \to 0$. Therefore, $\widehat{\beta_2}$ converges to β_2.

(iii) **Best Linear Unbiased Estimator (BLUE)**

The OLS estimators are the best linear unbiased estimators (BLUE). This property is also called the '**Gauss-Markov theorem'**. The theorem states that 'given the assumptions of classical linear regression models (CLRM), the least squares estimators have the minimum variance in the class of all linear unbiased estimators'. More specifically, the OLS estimator of $\widehat{\beta_2}$ is said to be BLUE of β_2. The OLS estimator $\widehat{\beta_2}$ has the following properties:

(i) Linear: The OLS estimator is a linear function of the dependent variable. As you know, $\widehat{\beta_2} = \frac{\sum x_i y_i}{\sum x_i^2}$. We can present it as

$\widehat{\beta_2} = \sum c_i y_i$ … (4.54)

where $c_i = \frac{x_i}{\sum x_i^2}$.

(ii) Unbiased: The OLS estimator is unbiased. We have proved in equation (4.52) that $\widehat{\beta_2}$ is unbiased estimator of β_2.

(iii) Efficient: The OLS estimator is efficient in the sense that in the class of all linear unbiased estimators, it has the minimum or the least variance. In order prove it, we proceed as follows:

Let us consider any linear unbiased estimator b_2 such that

$b_2 = \sum d_i \, y_i$ … (4.55)

$E(b_2) = \sum d_i \, E(y_i) = \sum d_i \, E(\beta_2 x_i) = \beta_2 \sum d_i x_i$

For b_2 to be unbiased, $E(b_2) = \beta_2$. For this, we should have

$\sum d_i x_i = 1$ … (4.56)

Since $b_2 = \sum d_i \, y_i$, we should have

$var(b_2) = \sum d_i^2 \, \sigma^2$ … (4.57)

Our objective is to find out the value of d_i for which $\sum d_i^2 \sigma^2$ is minimum. Also, it should fulfil the ondition that $\sum d_i x_i = 1$. Since σ^2 is a constant, we need to minimize the following Lagrangian function:

$$\sum d_i^2 - \lambda(\sum d_i x_i - 1) \qquad \ldots (4.58)$$

where λ is the Langrangian multiplier.

If we differentiate equation (4.58) with respect to d_i, and equate it to zero, we obtain

$$2d_i - \lambda x_i = 0$$

Or,

$$d_i = \frac{\lambda}{2} x_i \qquad \ldots (4.59)$$

Multiplying both sides of equation (4.59) by x_i and summing over the sample, we obtain

$$\sum d_i x_i = \frac{\lambda}{2} \sum x_i^2 \qquad \ldots (4.60)$$

Recall from (4.56) that $\sum d_i x_i$. Hence, $\frac{\lambda}{2} \sum x_i^2 = 1$. Therefore, $\lambda = \frac{2}{\sum x_i^2}$. Substituting this value of λ in (4.59), we obtain

$$d_i = \frac{2}{\sum x_i^2} \cdot \frac{1}{2} x_i = \frac{x_i}{\sum x_i^2} \qquad \ldots (4.61)$$

Therefore, the minimum variance of b_2 is achieved when (see equation (4.55)),

$$b_2 = \sum d_i y_i = \sum \frac{x_i}{\sum x_i^2} y_i = \frac{\sum x_i y_i}{\sum x_i^2} \qquad \ldots (4.62)$$

Compare equation (4.62) with the OLS estimator given at equation (4.14). Both are the same. Thus, the OLS estimator has the minimum variance among the linear unbaised estimators; and it is BLUE.

The Gauss-Markov theorem holds only if the assumptions of CLRM are satisfied. This ensures that we cannot get any other estimator with variance lower than that of the OLS estimator. A remarkable feature of the theorem is that it makes no assumption about the distribution of the error term u_i and Y_i. Any violation of the CLRM assumptions leads to invalidity of the theorem. For instance, if we have a regression model which is non-linear in parameters, we may obtain estimators which are better than OLS estimators. Another example is the case of violation of the homoscedastic assumption. Under such situations, the estimators may be unbiased and consistent, but the variance will not be the minimum even in the class of linear estimators. It is important to note that the statistical properties of the OLS estimator are valid, irrespective of the sample size.

Check Your Progress 2

1) State the variance of the OLS estimators.

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2) State the numerical properties of the OLS estimators.

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3) State the statistical properties of the OLS estimators.

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4) State the central features of Gauss Markov Theorem.

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5) Using the data given for question 2 in CYP 1, calculate the estimator of the error variance σ^2.

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4.5 GOODNESS OF FIT

Let us now examine how we can ascertain the 'goodness of fit' of the estimated regression line. A perfect fit would be obtained if all the observations were on the regression line. But this is a rare possibility. We normally get a distribution in such a way that some observations are below, and some observations are above the regression line. The distance between the actual value Y from the expected value $\hat{Y}$ will have both positive and negative values. Thus, the residual ($\hat{u}_i$) can be either positive or negative. What we require is that the value of this residual $\hat{u}_i$ is as small as possible. A measure of how well the sample regression line fits the data is given by the 'coefficient of determination' (r^2). Note that the notation r^2 is usually used for bi-variate regressions, while R^2 is used for multiple regression. How do we compute r^2? Let us once again consider our SRF as in equation (4.3), i.e., $Y_i = \widehat{Y_i} + \widehat{u_i}$. In deviation form, this can be written as:

$$y_i = \widehat{y_i} + \widehat{u_i} \qquad \text{... (4.63)}$$

Squaring (4.30) on both sides and summing up we get:

$$\sum y_i^2 = \sum \hat{y}_i^2 + \sum \hat{u}_i^2 + 2\sum \hat{y}_i\, \hat{u}_i$$

$$= \sum \hat{y}_i^2 + \sum \hat{u}_i^2 \qquad (\text{since} \sum \hat{y}_i\, \hat{u}_i = 0)$$

$$= \widehat{\beta_2}^2 \sum {x_i}^2 + \sum \hat{u}_i^2 \qquad \text{... (4.64)}$$

Equation (4.64) contains the following sum of squares:

(i) $\sum y_i^2$ represents the total variation of Y values about the mean. This is termed as the 'total sum of squares (TSS)'.

(ii) $\widehat{\beta_2}^2 \sum {x_i}^2$ represents the variation of estimated Y values about their mean. This is termed as the 'explained sum of squares (ESS)' or variation due to regression, or due to the explanatory variables.

(iii) $\sum \widehat{u_i}^2$ represents the residual or the unexplained variation of Y and is termed as the 'residual sum of squares (RSS)'.

We can therefore write (4.64) in terms of the sum of the squares as:

$$\text{TSS} = \text{ESS} + \text{RSS} \qquad \text{... (4.65)}$$

The total variations in Y (about its mean) is thus segregated into two parts: one part of the variation attributed to the regression and the second part to the random error. Dividing (4.65) by TSS on both sides we get:

$$1 = \frac{ESS}{TSS} + \frac{RSS}{TSS} \qquad \text{... (4.66)}$$

$$1 = \frac{\sum(\hat{Y}_i - \bar{Y})^2}{\sum(Y_i - \bar{Y})^2} + \frac{\sum \widehat{u_i}^2}{\sum(Y_i - \bar{Y})^2} \qquad \text{... (4.67)}$$

From (4.63), r^2 can be defined as:

$$r^2 = \frac{\sum(\hat{Y}_i - \bar{Y})^2}{\sum(Y_i - \bar{Y})^2} = \frac{ESS}{TSS} \qquad \text{... (4.68)}$$

Therefore, equation (4.67) can also be written as:

$$r^2 = 1 - \frac{\sum \widehat{u_i}^2}{\sum(Y_i - \bar{Y})^2}$$

$$= 1 - \frac{RSS}{TSS} \qquad \text{... (4.69)}$$

Alternatively, $r^2 = \frac{ESS}{TSS} = \frac{\sum(\hat{y}_i)^2}{\sum y_i^2} = \frac{\widehat{\beta_2}^2 \sum {x_i}^2}{\sum y_i^2}$

$$= \widehat{\beta_2}^2 \left(\frac{\sum x_i^2}{\sum y_i^2}\right) \quad \text{... (4.70)}$$

Dividing both the numerator and the denominator of the RHS of (4.70) by the sample size *n* we get:

$$r^2 = \widehat{\beta_2}^2 \left(\frac{S_x^2}{S_y^2}\right) \quad \text{... (4.71)}$$

where S_x^2 and S_y^2 are the sample variances of *X* and *Y* respectively. Substituting for $\widehat{\beta_2}$ from (4.14), equation (4.70) can be written as:

$$r^2 = \frac{\left(\sum x_i y_i\right)^2}{\sum x_i^2 \sum y_i^2} \quad \text{... (4.72)}$$

Therefore, as a formal definition of r^2, we can say that r^2 is the measure of the proportion of total variation in *Y* that is explained by *X* in the regression model. It provides the magnitude or the extent to which variation in the independent variable can affect the variation in the dependent variable. The value of r^2 is always non-negative and ranges between 0 to 1, that is, $0 \leq r^2 \leq 1$. A value of 1 would mean a perfect fit, i.e., $\widehat{Y_i} = Y_i$, whereas $r^2 = 0$ means that there is absolutely no relationship between the dependent and the explanatory variable.

4.6 TESTING OF HYPOTHESIS

The main idea of hypothesis testing is to examine if the sample value is compatible (i.e., sufficiently close) to the population value. For example, let us say that the theory or previous experience says that the true slope coefficient of β_2 is zero. But from the sample, we obtain the estimate as $\widehat{\beta_2} = 0.674$. Now, we need to test whether the estimate obtained is consistent with the hypothesis. In case it is consistent, we do not reject the hypothesis. Here, the hypothesis that β_2 is zero is called as the 'null hypothesis or H_o'. This is compared against the alternative hypothesis H_1 that β_2 is different from zero. The alternative hypothesis may be simple like H_1: $\beta_2 \neq 0$.

Hypothesis testing requires developing rules or procedures to decide whether to reject or not-reject the null hypothesis. There are two common approaches to hypothesis testing: (i) confidence interval approach, and (ii) test-of-significance approach. Both the approaches are based on the fact that the variable of our concern has some probability distribution. Hypothesis testing involves making statements or claims about the value of the parameter given such a distribution. Here we first focus on the 'test-of-significance approach'.

The test-of-significance approach refers to a procedure for testing a null hypothesis (H_o) using the sample results. The procedure involves obtaining the test statistic (estimator) and the distribution of such a statistic under the null hypothesis. Based on the value of the test statistic thus obtained, the decision to accept or reject the null hypothesis is made. Under the assumption of normal distribution, we can write the test statistic as:

$t = \frac{\widehat{\beta_2} - \beta_2}{se\,(\widehat{\beta_2})}$ ... (4.73)

Equation (4.70) follows *t*-distribution with $(n-2)$ degrees of freedom in the simple regression model. If the value of β_2 is specified under the null hypothesis, we can compute the value of *t*-statistic using the sample data. Since this statistic (*t*) follows *t*-distribution, its confidence interval can be stated as:

$Pr\left[- t\alpha_{/2} \leq \frac{\widehat{\beta_2} - \beta_2{}^*}{se\,(\widehat{\beta_2})} \leq t\alpha_{/2}\right] = (1 - \alpha)$ ... (4.74)

where β_2^* is the value of β_2 under the null hypothesis and $-t\alpha_{/2}$ and $t\alpha_{/2}$ are the critical *t*-values (from the *t*–Table) for $\frac{\alpha}{2}$ level of significance with $(n-2)$ degrees of freedom. Rearranging (4.74) we get:

$Pr\left[\beta_2^* - t\alpha_{/2} se(\widehat{\beta_2}) \leq \widehat{\beta_2} \leq \beta_2^* + t\alpha_{/2} se(\widehat{\beta_2})\right] = (1 - \alpha)$ ... (4.75)

Equation (4.75) gives the interval in which $\widehat{\beta_2}$ will lie with a probability of $(1 - \alpha)$ if $\beta_2 = \beta_2^*$. The 100(1– α) % confidence interval is termed as the 'region of acceptance' of the null hypothesis. The region beyond this is termed as the 'region of rejection' or the 'critical region'. In the confidence interval approach, a true value of β_2 is hypothesized and then it is examined if the estimated $\widehat{\beta_2}$ lies within the region of acceptance or not.

Example 4.1

Let us consider an example where it is hypothesized that the true value of $\beta_2 = 0.3$, i.e., *Ho*: $\beta_2 = 0.3$. The *H0* is tested against the alternative hypothesis *H1*: $\beta_2 \neq 0.3$. Let $\widehat{\beta_2} = 0.5091$ and $se(\widehat{\beta_2}) = 0.0357$ with *d.f.* = 8. Let us assume that α = 5 % so that $t\alpha_{/2} = 2.306$. Now using (4.75) we have the confidence interval for $\widehat{\beta_2}$ as:

$Pr[0.2177 \leq \widehat{\beta_2} \leq 0.3823] = 0.95$ ... (4.76)

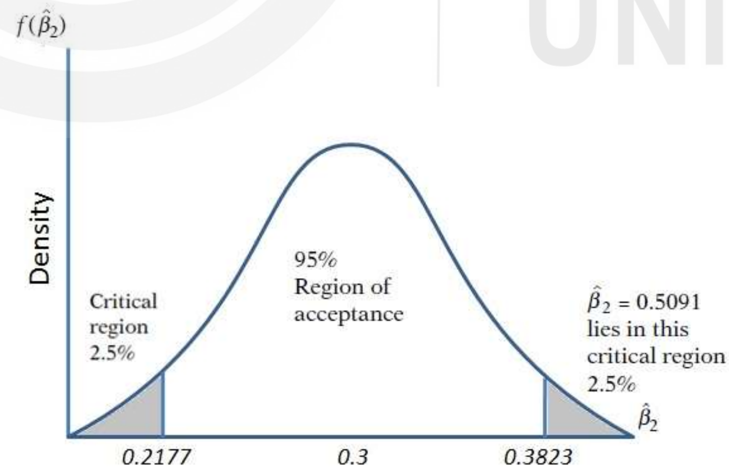

Fig.4.1: The 95% Confidence Interval for $\widehat{\beta_2}$ (under *H0*: β_2 = 0.3)

Equation (4.76) is represented in Fig. 4.1. As the estimated $\widehat{\beta_2}$ lies in the critical region, we reject the null *Ho*: $\beta_2 = 0.3$. This can also be verified empirically by calculating the *t*-value using (4.73) as:

$$t = \frac{0.5091 - 0.3}{0.0357} = 5.86$$

This is again greater than 2.306, leading to the rejection of H_o: $\beta_2 = 0.3$. From (4.70), it is evident that if $\widehat{\beta_2}$ is equal to the hypothesized β_2, the *t*-value will be 0. As the difference between the two increases, the $|t|$ will become larger. Note that the *t*-value can be both positive and negative. Thus, large $|t|$ is an indication of rejection of null hypothesis.

When the alternative hypothesis is like H_1: $\beta_2 \neq 0.3$, it is a case of two-tailed or two-sided test. This is because we consider the two extreme tails of the probability distribution as the region of rejection (as in Fig. 4.1). If we had the H_0 as $\beta_2 \leq 0.3$ and H_1 as $\beta_2 > 0.3$, we use one sided (or, one-tail) test (or right-tail test) as shown in Fig.4.2. The test procedure is the same as discussed before. The only difference is that in one-tailed test, the upper critical value corresponds to $t_\alpha = t_{0.05}$ or 5% level. The lower-tail is no longer considered in this case. The choice of two-tailed or one-tailed test depends on the nature of the alternative hypothesis formulated based on a-priori considerations. The decision on the type of t-test for hypothesis testing is summarized in Table 4.1.

4.7 FORECASTING

Regression is useful for forecasting the values of *Y* corresponding to the given *X* values. This may be the conditional mean value of *Y* for the selected value of *X*, say X_0, representing a point on the regression line itself. This is called 'mean forecasting'. Alternatively, it can be the forecasting of an individual value of *Y* corresponding to X_o. This is called 'individual forecasting'.

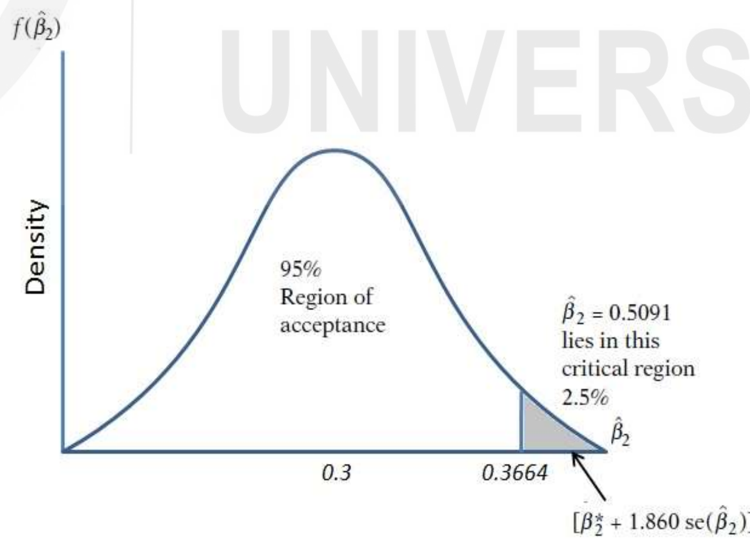

Fig.4.2: One-Tailed Test of Significance.

Table 4.1: Decision Rules for *t*-Test

Type of Hypothesis	H_0: Null hypothesis	H_1: Alternative Hypothesis	Decision – Reject H_0 if
Two-tail	$\beta_2 = \beta_2^*$	$\beta_2 \neq \beta_2^*$	$\lvert t\rvert > t_{\alpha/2}, df$
Right-tail	$\beta_2 \leq \beta_2^*$	$\beta_2 > \beta_2^*$	$t > t_\alpha, df$
Left-tail	$\beta_2 \geq \beta_2^*$	$\beta_2 < \beta_2^*$	$t < -t_\alpha, df$

Notes: β_2^* is the hypothesized value of β_2 ; $|t|$ is the absolute value of *t*, $t_{\alpha/2}$ and t_α are the critical values of α at $\alpha/2$ level of significance; df is the degrees of freedom.

4.7.1 Mean Forecasting

Suppose we want to predict the conditional mean of *Y* given $X_o = 100$, i.e., $E(Y|X_o = 100)$. Such a point estimate of the mean prediction is given as:

$$\widehat{Y_o} = \widehat{\beta_1} + \widehat{\beta_2} X_o \qquad \dots (4.77)$$

Since $\widehat{Y_o}$ is the estimator, the difference between this estimator and the true value indicates the predicted or the forecasted error. In this context, the variance of the estimator is given by:

$$var(\widehat{Y_o}) = \sigma^2 \left[\frac{1}{n} + \frac{(X_o - \overline{X})^2}{\sum x_i^2}\right] \qquad \dots (4.78)$$

We can compute the *t*-value by substituting the unknown σ^2 with its unbiased estimator $\widehat{\sigma^2}$. Then the statistic:

$t =$ follows *t*–distribution with (*n* -2) *df.* Its confidence interval is given by:

$$Pr\left[\widehat{\beta_1} + \widehat{\beta_2}Xo - t_{\alpha/2} se(\widehat{Y_o}) \leq \beta_1 + \beta_2 X_o \leq \widehat{\beta_1} + \widehat{\beta_2} X_0 + t_{\alpha/2} se(\widehat{Y_o})] = 1 - \alpha\right] \dots (4.79)$$

4.7.2 Individual Forecasting

Suppose we are interested in forecasting an individual *Y* value, Y_o, for the corresponding given value of X_o. Then, the variance of the best linear unbiased estimator of Y_0 is given by:

$$var(Y_o - \widehat{Y_o}) = E\left(\left(Y_o - \widehat{Y_o}\right)^2\right) = \sigma^2 \left[1 + \frac{1}{n} + \frac{(X_o - \overline{X})^2}{\sum x_i^2}\right] \qquad \dots (4.80)$$

Substituting the unknown σ^2 with the unbiased estimators $\widehat{\sigma^2}$ [by using the square root of (4.80) for *se* of $\widehat{Y_o}$], the statistic in (4.73) can be computed. Note that: (i) for each value of *X* plotted, we get a confidence band, and that, (ii) the 95% confidence interval of any $\widehat{Y_o}$ is always wider than the confidence interval for the mean value Y_o. In other words, the width of the confidence band is smallest when $X_o = \overline{X}$ and it gets wider as X_o moves away from $\overline{X}$. This means that the predictive or forecasting capacity of an X_o decreases considerably as it departs more and more from $\overline{X}$. We therefore need to be cautious in extrapolating the regression line for the purpose of predicting $E(Y|X_o)$.

Check Your Progress 3

1) Look into question 2 of CYP 1 again. Compute the value of r^2. Also, interpret the calculated value of r^2.

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2) Using the value of r^2 computed above, compute the test statistic for testing the null hypothesis H_o: $\beta_2 = 0$. Take the alternative hypothesis as H_1: $\beta_2 \neq 0$. Test at 5% significance level ($\alpha = 0.05$). Interpret the test outcome.

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3) While predicting the $E(Y|X_o)$, i.e., Y_o for the given X_o representing a point on the regression line, why does the predictive capacity of X_o decrease as the distance of X_o from $\overline{X}$ increases?

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4.7 LET US SUM UP

In this Unit we began with the specification of two-variable regression and its estimation methods. There are three methods of estimation: method of least squares, method of moments, and method of maximum likelihood. We specified the classical assumptions regarding the stochastic error term. The algebraic properties and the statistical properties of the regression model are spelt out. The explanatory power or goodness of fit of a regression model is given by the coefficient of determination. Subsequently, we discussed the procedure of testing a hypothesis. Many related concepts like coefficient of determination, level of significance, critical region and confidence interval are also explained in the unit.